ABSTRACT 1
Problem Description 2
Background and Community Impact 3
Algorithm and Decision Rules 3
Pseudocode 5
Flowchart Outline 6
Student Information Input 6
Validating the Attendance Data 7
Declaring the Alert Variable and Attendance Rate 7
Checking the Attendance Rate 8
Displaying the Results 8
Ending the Program 8
Sample Calculation and Expected Output 8
SDG Alignment 9
Open-Source Justification 9
Digital Public Goods Considerations 9
Limitations and Assumptions 9
Conclusion 9
References 10

ABSTRACT

This project presents the logical design of a School Attendance Monitoring and Early Support Alert System for secondary schools in Sierra Leone. The proposed system records student attendance, calculates attendance rates over a defined period, identifies repeated absences and produces an alert for authorised school staff when a student may need follow-up. It applies computational-thinking concepts by dividing the task into data entry, validation, calculation, decision-making, reporting and record updating. The report includes variables and data types, an algorithm, pseudocode, a flowchart outline, modular functions, decision and repetition structures, sample calculations, test cases and considerations for open-source software and digital public goods. The system is intended to support—not replace—teachers, school administrators and communication with families. It is a conceptual assignment design, not an existing government or school platform, and attendance alerts should be reviewed by a person before action is taken.

Problem Description

Regular school attendance supports continuity in learning, classroom participation and completion of schoolwork. When attendance is recorded only on paper or checked inconsistently, it may be difficult for school staff to identify patterns of repeated absence quickly. Records can also be time-consuming to summarise across a week or month.

The problem selected for this project is the organisation and review of attendance records. The proposed system provides a simple logical process for entering attendance, validating records, calculating attendance rates and flagging patterns that may require follow-up. It does not assume that every absence is caused by a student's lack of interest. Absence may be related to illness, transport, family responsibilities, financial difficulty, safety or other circumstances, so an alert should prompt supportive inquiry rather than punishment.

Background and Community Impact

Schools need dependable records to understand attendance patterns and plan appropriate support. A basic digital or computer-based process could help teachers and administrators review records consistently, particularly when a class contains many students. The project is designed for possible adaptation to secondary schools in Sierra Leone, including schools that may begin with simple spreadsheet-like records before adopting a fuller system.

Potential users include class teachers, attendance officers, school administrators and authorised staff responsible for contacting families. Students and families may benefit if repeated absence is noticed early and the school can investigate barriers and provide appropriate support. The system cannot itself remove barriers such as transport costs, illness or lack of learning resources; it can only help make patterns more visible.

Aim: To design a logical system for recording, summarising and reviewing student attendance so that repeated absences can be identified for supportive follow-up.

- Record attendance for each student on each school day.
- Validate student identifiers and attendance values before saving a record.
- Calculate attendance totals and rates for a selected period.
- Apply transparent rules to identify attendance patterns that may need review.
- Generate a class summary and a list of students requiring follow-up.
- Demonstrate variables, data types, input, processing, output, decisions, loops and modular functions.
- Consider privacy, fairness and open-source design in a possible future implementation.

Algorithm and Decision Rules

For this demonstration, the attendance rate is calculated as:

attendanceRate = (daysPresent / daysHeld) × 100

The calculation is performed only when daysHeld is greater than zero. In this example, an alert is recommended when attendanceRate is below 85%, or when a student has been absent for three consecutive school sessions. These thresholds are illustrative project assumptions, not an official Ministry of Basic and Senior Secondary Education rule. A school should approve any threshold before use.

- If the student ID is missing or attendance status is not an allowed value, reject the entry and request correction.
- If daysHeld equals zero, do not calculate a percentage.
- If attendanceRate is below 85%, set alertFlag to TRUE.
- Otherwise, set alertFlag to FALSE.
- A staff member reviews each alert and checks the context before contacting the student or family.

```
BEGIN
    DECLARE StudentID AS INTEGER
    DECLARE NOSTUDENTS AS INTEGER
    DECLARE STUDENTCOUNTER AS INTEGER
    ASSIGN STUDENT ID = 0
    DECLARE AttendanceRate AS REAL
    DECLARE Choice AS STRING
    OUTPUT "SCHOOL ATTENDANCE MONITORING SYSTEM"
    OUTPUT "Enter Student Name:"
    INPUT StudentName
        OUTPUT "Enter Student ID"
    INPUT STUDENTID
    OUTPUT "Enter class"
    INPUT CLASS
OUTPUT “ATTENDANCE STATUS? ”
INPUT ATTENDANCE
STUDENTCOUNTER = STUDENTCOUNTER + 1
DECLARE DAYSHELD = 60
INTEGER DAYSPRESENT
OUTPUT “INPUT DAYS PRESENT”
INPUT DAYSPRESENT
SET DAYSHELD<=0
IF DAYSHELD<=0 THEN
OUTPUT “CANNOT CALCULATE ATTENDANCE RATE”
IF DAYSHELD >=0 THEN
       SET AttendanceRate = (DAYSPRESENT / DAYSHELD) * 100
    IF AttendanceRate < 85 OR THEN
       OUTPUT "ATTENDANCE ALERT: YES" AND STUDENT STATUS
    ELSE
       OUTPUT "ATTENDANCE SATISFACTORY."  END IF
```

Pseudocode

```
BEGIN
    INPUT className, schoolDate
    INPUT numberOfStudents
    SET studentCounter = 1

    WHILE studentCounter <= numberOfStudents
        INPUT studentID, studentName, attendanceStatus

        IF studentID is empty THEN
            OUTPUT "Invalid record: student ID is required"
        ELSE IF attendanceStatus NOT IN
                ["PRESENT", "ABSENT", "LATE", "AUTHORISED"] THEN
            OUTPUT "Invalid status: enter an approved value"
        ELSE
            SAVE attendance record

            IF attendanceStatus = "PRESENT" OR
               attendanceStatus = "LATE" THEN
                UPDATE attendance totals according to school policy
            END IF

            OUTPUT "Attendance record accepted"
        END IF

        SET studentCounter = studentCounter + 1
    END WHILE

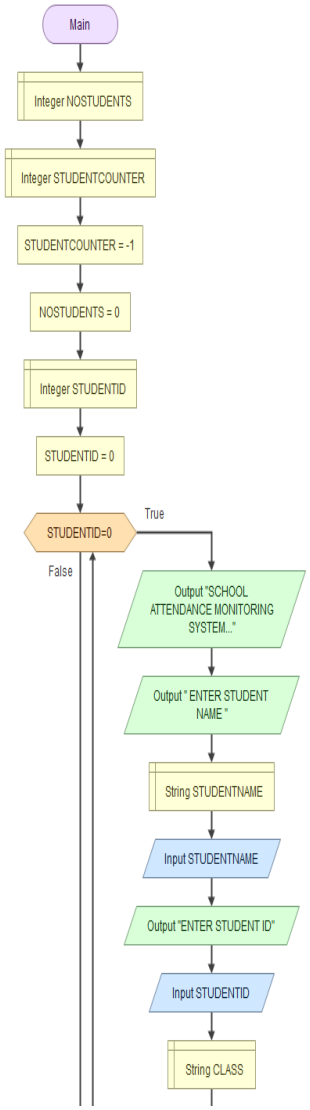
    INPUT daysHeld
    IF daysHeld <= 0 THEN
        OUTPUT "Cannot calculate attendance rate"
    ELSE
        FOR each student in selected period
            CALCULATE attendanceRate =
                (daysPresent / daysHeld) * 100

            IF attendanceRate < 85 OR absenceStreak >= 3 THEN
                SET alertFlag = TRUE
                OUTPUT studentID, attendanceRate, "Review required"
            ELSE
                SET alertFlag = FALSE
            END IF

OUTPUT class attendance summary
END
```

Flowchart Outline

Recreating the following logic in Flowgorithm using Start/End, Input, Output, Assignment, If, While and For symbols:



Main Process and Variable Initialization

The program begins at the **Main** symbol. It declares the NOSTUDENTS and STUDENTCOUNTER variables as integers.

- STUDENTCOUNTER is initialized to -1.
- NOSTUDENTS is initialized to 0.
- STUDENTID is declared as an integer and initialized to 0.

The program then checks whether STUDENTID = 0. This condition determines

Student Information Input

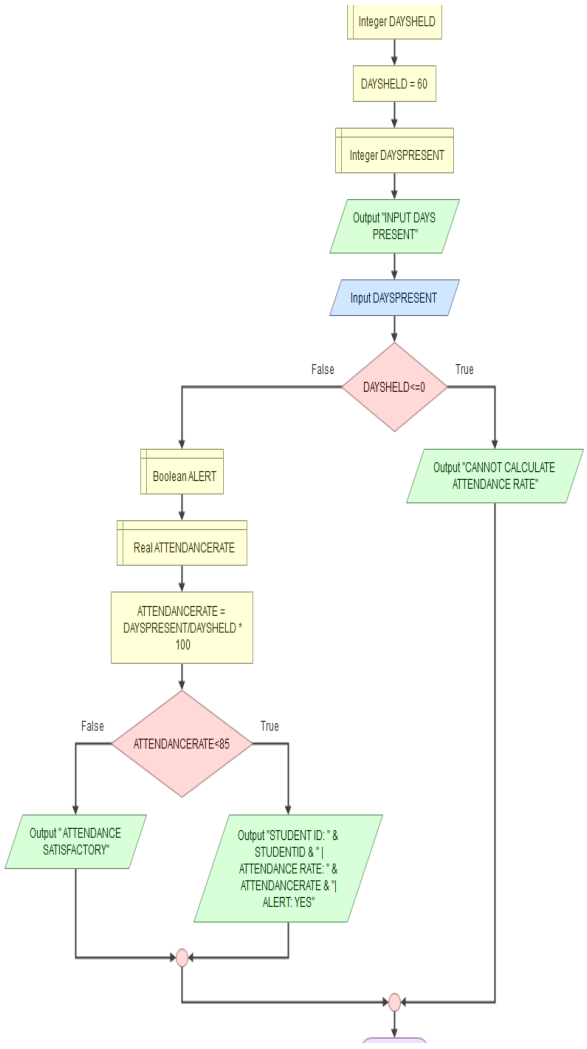
When the condition STUDENTID = 0 is true, the program displays the title **School Attendance Monitoring System** and asks the user to enter the student's name.

The program then:

- Declares STUDENTNAME as a String.
- Accepts the student's name as input.
- Requests and accepts the Student ID.
- Declares CLASS as a String and accepts the student's class.
- Declares DATE as an Integer and accepts the date.
- Requests the student's attendance status.
- Declares ATTENDANCE as a String and accepts the attendance status.

After collecting the information, the program increases STUDENTCOUNTER by one.

Purpose: This process collects the student's basic information and attendance status for recording.



Validating the Attendance Data

The program checks the condition:

DAYSHELD <= 0

- **If True:** The program displays the message *"Cannot calculate attendance rate"* because the total number of days must be greater than zero.
- **If False:** The program proceeds to calculate the attendance rate.

Purpose: This decision prevents the program from dividing by zero or using a negative number of school days.

Declaring the Alert Variable and Attendance Rate

When DAYSHELD is greater than zero, the program declares:

- ALERT as a Boolean variable.
- ATTENDANCERATE as a Real variable.

It then calculates the attendance rate using the formula:

Attendance rate= days present/daysheld*100

For example, if a student attends 45 out of 60 school days:

45/60*100=75\%

The student's attendance rate is therefore 75%.

Purpose: This process converts the student's attendance information into a percentage that can be used to evaluate attendance.

Checking the Attendance Rate

The program evaluates the condition:

ATTENDANCERATE < 85

This decision determines whether the student's attendance rate is below 85%.

- **If True:** The program displays the Student ID, attendance rate, and an alert indicating that the student requires attention.
- **If False:** The program displays *"Attendance Satisfactory"*.

For example:

- An attendance rate of 75% triggers an alert.
- An attendance rate of 90% is classified as satisfactory.
- An attendance rate of exactly 85% is classified as satisfactory.

Purpose: This process identifies students whose attendance rates fall below the project's chosen threshold of 85%.

Displaying the Results

When an alert is triggered, the program displays the Student ID, attendance rate, and an indication that an alert is required.

When the attendance rate is 85% or higher, it displays the message *"Attendance Satisfactory"*.

If the total days held are invalid, the program displays an error message instead.

Purpose: This process communicates the program's results to the user so that attendance concerns can be identified.

Ending the Program

After displaying the relevant result or error message, the program's execution paths join and proceed to the **End** symbol.

The program then terminates.

Sample Calculation and Expected Output

Illustrative example: A student has 17 present sessions out of 20 school sessions during the selected period.

attendanceRate = (17 / 20) × 100 = 85%

At exactly 85%, the example rate rule does not trigger an alert because the rule is strictly below 85%.

Student ID: ST-024
Attendance rate: 85%
Alert: YES Reason: attendance rate low, staff review required

SDG Alignment

The primary alignment is Sustainable Development Goal 4: Quality Education. Reliable attendance information can help schools notice participation barriers and consider support that helps students remain engaged in education. The system contributes indirectly: it does not itself provide teachers, transport, learning materials or other resources. Its purpose is to improve the visibility and review of attendance patterns.

Open-Source Justification

A future implementation could use an approved open-source licence so developers and schools can inspect, adapt and improve the software under the licence terms. Open-source development could make the logic easier to review and allow adaptation to different school contexts. Documentation, accessible data formats and a simple interface would be important. This report proposes an open-source approach; it does not claim that a finished application already exists.

Digital Public Goods Considerations

A future version could be designed with digital public goods principles in mind, including relevance to the SDGs, open licensing, documentation, platform independence, privacy and security. Attendance records are sensitive information about children, so the system should restrict access to authorised school staff, collect only necessary data, protect backups and avoid publishing identifiable student information. A full legal and safeguarding review would be required before any

Limitations and Assumptions

- The 85% threshold and three-session rule are illustrative assumptions, not official policy.
- Attendance status definitions and treatment of lateness or authorised absence must be set by the school.
- Incorrect or incomplete records can produce misleading summaries.
- An alert does not explain why a student was absent and must not be treated as proof of wrongdoing.
- A real system needs authentication, access controls, backup, correction logs and a data-retention policy.
- The project does not automatically contact families or make disciplinary decisions.

Conclusion

The School Attendance Monitoring and Early Support Alert System demonstrates how programming logic can address a practical education-related challenge. Input validation, variables, arithmetic calculations, Boolean conditions, decisions, repetition and modular functions support consistent attendance recording and summarisation. By flagging patterns for human review, the system could help schools identify students who may need support. Its usefulness would depend on accurate data, privacy safeguards, school-approved rules and careful interpretation of each student's circumstances.

References

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3. Open Source Initiative (n.d.). The Open Source Definition. https://opensource.org/osd
4. Digital Public Goods Alliance (n.d.). Digital Public Goods Standard. https://www.digitalpublicgoods.net/standard